

MARHAM, United Kingdom

WMO: 034820

Lat: 52.6514N

Lon: 0.5661E

Elev: 75

StdP: 14.66

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.7	27.3	21.0	15.8	28.3	23.6	17.9	30.3	33.0	49.2	29.9	48.9	4.5	20	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	80.7	66.0	77.0	64.3	73.8	62.8	67.7	77.3	65.8	74.4	64.1	71.5	10.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	89.5	71.3	62.5	84.8	69.1	61.1	80.7	67.6	32.0	77.5	30.6	74.7	29.4	71.7	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.4	24.3	21.6	DB	18.7	87.4	4.3	3.0	15.6	89.5	13.1	91.2	10.6	92.9	7.5	95.0	(4)
			WB	17.9	71.5	4.4	2.8	14.8	73.5	12.2	75.2	9.8	76.7	6.6	78.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	39.4	40.3	43.5	48.2	53.6	58.7	62.9	62.8	58.0	52.0	44.5	40.2
	DBStd	10.11	5.87	5.91	5.50	5.59	5.62	5.07	4.83	4.78	4.64	5.50	5.71	6.54
	HDD50	1471	329	275	210	99	28	1	0	0	1	41	179	308
	HDD65	5413	792	692	666	505	355	199	100	102	216	403	614	769
	CDD50	1618	1	3	9	43	139	263	401	396	241	103	15	4
	CDD65	87	0	0	0	0	1	10	36	34	5	1	0	0
	CDH74	689	0	0	0	4	16	94	295	238	37	5	0	0
	CDH80	129	0	0	0	0	0	16	56	52	4	1	0	0
Wind	WSAvg	10.4	11.6	11.7	11.2	10.2	10.2	9.4	9.1	9.3	9.5	10.5	10.6	11.0
Precipitation	PrecAvg	26.6	2.3	1.8	1.6	1.7	2.0	2.2	2.5	2.6	2.3	2.6	2.6	2.4
	PrecMax	33.1	3.9	3.4	3.3	5.4	4.9	6.1	4.5	6.1	4.3	4.6	4.4	4.3
	PrecMin	18.8	0.6	0.5	0.2	0.0	0.5	0.6	1.1	0.2	0.6	0.7	0.8	1.1
	PrecStd	4.2	0.9	0.7	0.8	1.3	1.1	1.5	1.0	1.4	1.0	1.0	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	57.9	62.0	71.9	76.2	82.4	85.8	85.3	78.7	70.0	60.5	56.6
		MCWB	52.2	52.2	52.4	57.9	62.2	67.7	67.9	67.3	64.9	61.1	57.2	53.4
	2%	DB	52.9	53.9	58.5	65.2	71.9	76.2	80.7	80.3	73.2	65.2	57.6	54.4
		MCWB	50.2	49.9	51.5	54.9	60.7	64.1	66.0	66.2	62.7	59.3	54.6	51.9
	5%	DB	50.6	51.5	55.5	61.3	68.3	72.2	77.1	75.7	69.6	62.5	55.4	52.2
		MCWB	48.1	48.0	49.9	52.5	58.5	62.1	64.4	63.9	61.0	58.0	52.6	50.0
	10%	DB	48.4	49.3	52.9	58.2	64.4	68.9	73.6	72.3	66.5	60.3	53.2	50.1
		MCWB	46.2	46.5	48.2	50.8	56.2	59.9	62.8	62.3	59.2	56.5	50.8	47.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	53.0	54.7	59.4	65.0	69.4	70.0	69.9	67.3	62.8	57.8	54.1
		MCDB	54.7	56.2	59.5	68.8	73.0	80.4	81.8	82.3	75.3	68.2	59.8	55.9
	2%	WB	50.6	50.4	52.5	55.8	61.6	65.4	67.6	67.2	64.3	60.4	55.1	52.4
		MCDB	52.5	53.3	57.0	63.9	69.5	73.1	77.9	76.4	70.8	64.3	57.1	54.2
	5%	WB	48.4	48.5	50.6	53.7	59.3	63.1	65.7	65.4	62.3	58.8	53.0	50.2
		MCDB	50.3	51.0	54.8	59.9	66.8	70.1	74.8	73.7	67.7	61.7	55.0	52.2
	10%	WB	46.4	46.8	48.7	51.6	57.0	61.1	63.9	63.7	60.6	57.1	51.1	48.1
		MCDB	48.3	49.2	52.6	57.3	63.6	67.5	71.6	70.3	64.9	59.9	53.0	49.9
Mean Daily Temperature Range	5% DB	MDBR	9.8	11.2	13.7	15.9	16.3	16.0	16.9	16.8	15.8	13.3	11.0	9.5
		MCDBR	11.4	13.1	17.4	20.6	22.1	21.5	22.9	21.6	19.5	14.2	11.3	10.3
		MCWBR	10.4	10.0	11.6	11.8	12.7	11.6	10.5	9.6	10.5	9.5	9.6	9.5
	5% WB	MCDBR	11.5	12.2	16.0	19.2	20.5	19.1	20.9	19.4	17.2	13.0	11.0	10.2
		MCWBR	10.8	10.0	11.5	11.8	12.6	11.2	10.6	9.5	10.6	9.6	9.8	9.7
Clear-Sky Solar Irradiance	taub	0.326	0.343	0.377	0.402	0.405	0.410	0.410	0.398	0.377	0.357	0.330	0.316	
	taud	2.353	2.374	2.274	2.222	2.263	2.271	2.283	2.334	2.391	2.426	2.401	2.338	
	Ebn at Noon	208	242	255	263	268	267	265	262	253	233	205	188	
	Edh at Noon	19	25	34	40	40	40	39	35	30	24	18	16	
All-Sky Solar Radiation	RadAvg	226	443	802	1261	1540	1624	1556	1314	979	566	282	180	
	RadStd	22	65	85	131	117	123	158	92	81	49	29	21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page